

CATHOLIC JUNIOR COLLEGE
JC2 MID-YEAR EXAMINATIONS
Higher 1

**CANDIDATE
NAME**

CLASS

**INDEX
NUMBER**

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BIOLOGY

8875/02

Paper 2 Core Paper

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name on all the work you hand in.

Write in dark blue or black pen on both sides of the paper. **[PILOT FRIXION ERASABLE PENS ARE NOT ALLOWED]**

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

SECTION A]

Answer **all** questions.

SECTION B]

Answer any **one** question.

At the end of the examination, fasten all work securely together.

The number of marks is given in brackets [] at the end of each question or part of the question.

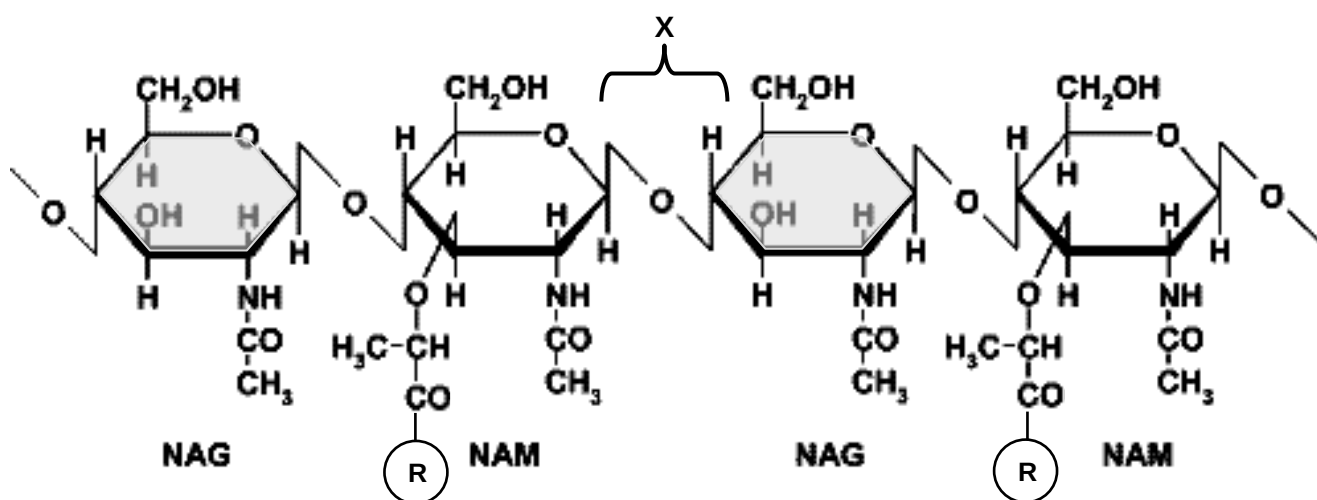
For Examiner's Use	
Section A	40
1 [7]	
2 [10]	
3 [12]	
4 [11]	
Section B	20
5 or 6	
TOTAL	60

Section A

Answer **all** the questions in this section.

- 1 The peptidoglycan cell wall of a prokaryote, *Escherichia coli*, is made up of two different sugars: N-acetylmuramate (NAM) and N-acetylglucosamine (NAG). NAM and NAG have different groups being attached to the C3 position.

Fig.1.1 shows the molecular structure of NAM and NAG.



Legend

(R) Short amino acid chain

Fig. 1.1

Fig.1.2 shows the structure of the peptidoglycan cell wall of *Escherichia coli*.

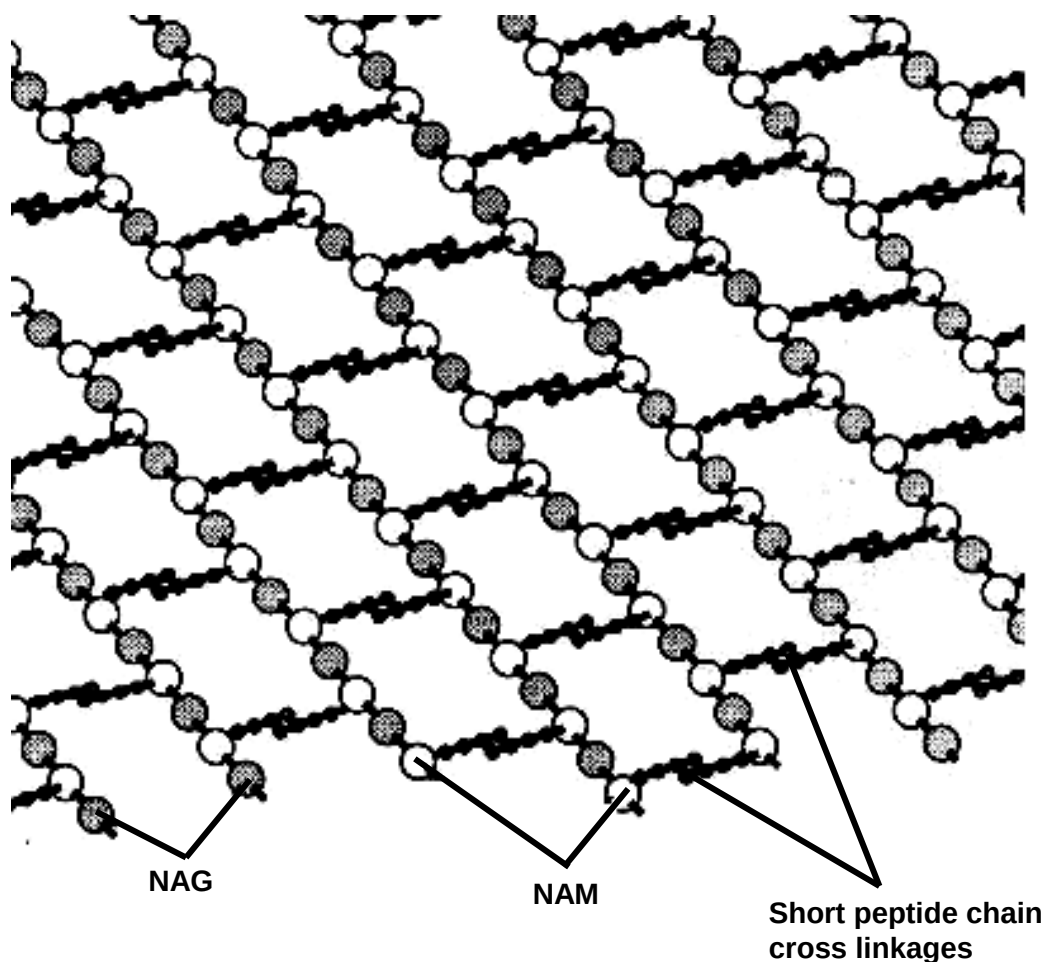


Fig 1.2

(a) With reference to Fig.1.1 and Fig. 1.2,

(i) name the bond denoted by **X**.

.....[1]

(ii) distinguish between bacterial peptidoglycan and cellulose found in the plant cell wall.

.....
.....
.....
.....
.....[2]

(iii) suggest why peptidoglycan associates with amino acids.

.....
.....
.....[1]

Lysozyme is considered antibacterial as it degrades polysaccharides that are found in the cell wall of many bacteria, including that of *Escherichia coli*. It is a globular protein with a deep cleft across a part of its surface. Six hexoses of the substrate can fit into this cleft.

Fig. 1.3 shows the action of part of lysozyme to the cell wall of *Escherichia coli*.

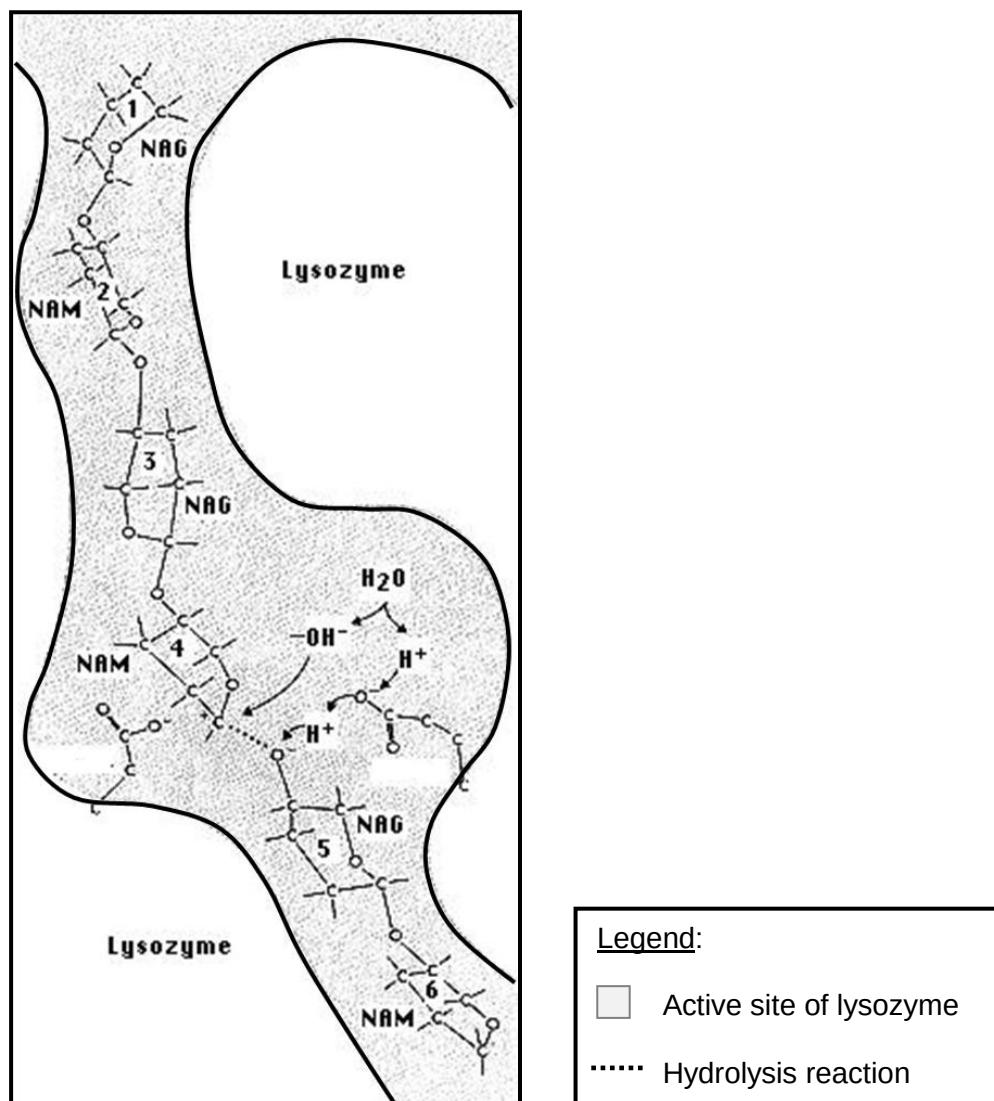


Fig.1.3

(b) With reference to Fig. 1.3, suggest how lysozyme is anti-bacterial.

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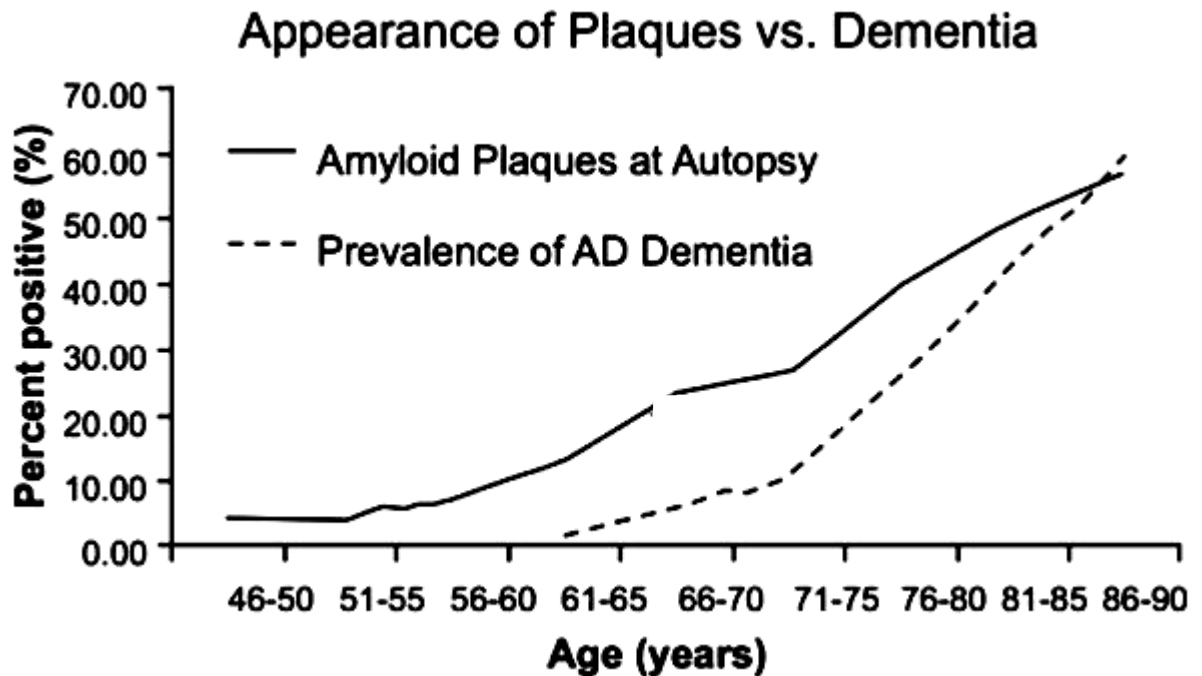
.....

.....[3]

[Total: 7]

- 2 (a) The *APP* gene is found in chromosome 21 and encodes for a protein known as amyloid precursor protein (APP). This protein is involved in synapse formation and development of neurons. The APP undergoes further processing to form soluble amyloid precursor protein (sAPP) and amyloid beta ($A\beta$). Studies have shown that $A\beta$ is likely to play a natural role in the ability of neurons to adapt and change over time (plasticity). However, when APP gene undergoes mutation, an overproduction of $A\beta$ will occur and they will be released from such mutated cells. Excess $A\beta$ will accumulate in the brain and aggregate to form clumps known as amyloid plaques.

In a large scale study, autopsies were carried out to determine the presence of amyloid plaques in people across different age groups (as indicated by percentage positive). This data was analyzed together with the prevalence of Alzheimer disease-related dementia (AD-related dementia) diagnosed in patients across different age groups (as indicated by percentage positive). The data is being presented in Fig. 2.1.



Source: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3220946/>

Fig. 2.1

- (i) With reference to Fig. 2.1, explain the relationship between amyloid plaques formation at autopsy to that of disease AD Dementia.

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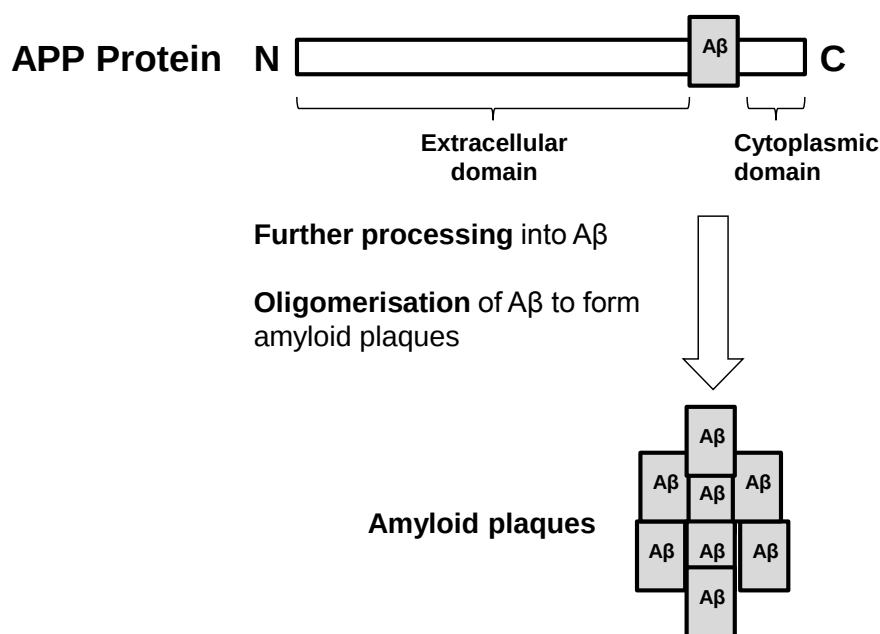
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..... [2]

Proteins with significantly different structures in various cells were found to have originated from the *APP* gene.



Modified from Source: <http://www.sciencedirect.com/science/article/pii/S0361923011001870>

Fig. 2.2

(ii) Describe and explain how such proteins could have arisen from a single gene.

.....

[3]

(iii) Suggest how excess Aβ can be removed within a cell.

.....

[1]

(b) Figure 2.1 shows a linear chromosome undergoing the first round of DNA replication.

A, B C and D represents the ends of the newly-synthesized DNA strand.

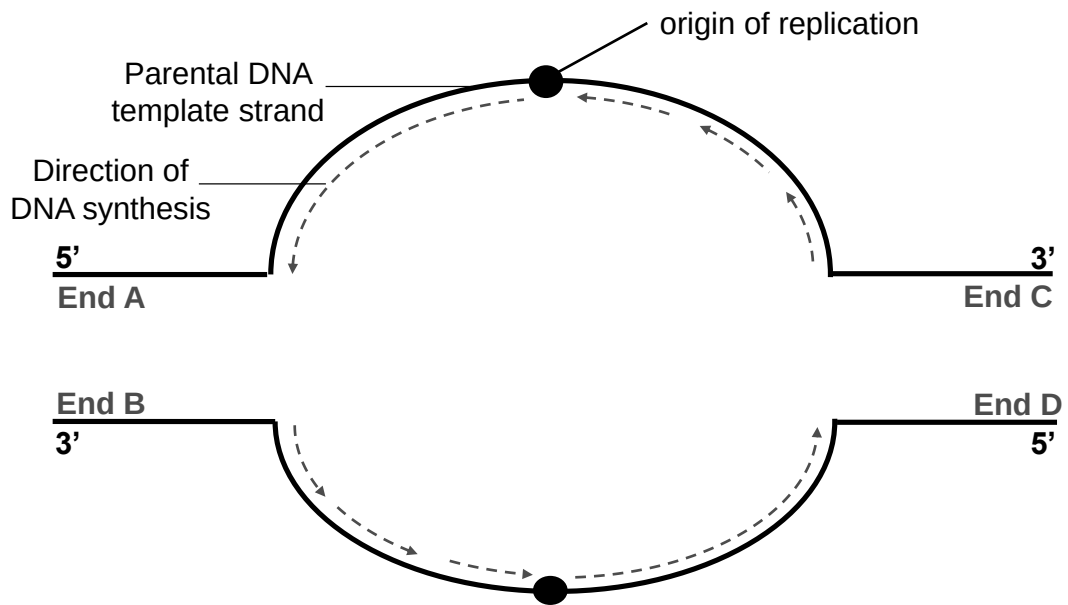


Fig. 2.3

(i) State which end(s) [A, B, C or D] of the newly-synthesised DNA strands will experience the end-replication problem after the first round of DNA replication has been completed.

.....
[1]

(ii) Suggest one modification that a researcher could do to the original parental duplex in Fig. 2.1 in order to completely remove the end-replication problem.

.....

[1]

(iii) Telomerase is an enzyme that is found in certain cell types and it helps to ameliorate the end-replication problem by lengthening the ends of the chromosomes, which are also known as telomeres.

Explain why the telomerase enzyme does not prevent the end-replication problem from occurring despite being able to lengthen the telomeres.

.....

[2]

[Total: 10]

- 3 (a)** The Manx gene determines the presence or absence of tails in cats. The Manx gene is controlled by Gene S which determines the development of the embryo cat's spine. The allele S^N codes for normal spine which results in long tail being form and S^A codes for abnormal, short spine. Cats which are homozygous recessive will not have their spines developed and so this genotype will die early in development. Manx cats are tailless with a heterozygous genotype, as shown in Fig. 3.1.



Fig. 3.1

A second gene which controls for the appearance of the coat is located on the X chromosome. Allele D codes for 'dappled' coat (D) while allele d codes for 'plain' coat (d).

A male Manx cat with dappled coat was mated on several occasions to the same female and a large number of offspring consisted of the following phenotypes, as summarised in Table 3.1 is observed.

Table 3.1

Tail Type	Gender	Coat colour	% no. of kittens
Long tailed	Male	Dappled	12.5
Long tailed	Female	Dappled	12.5
Manx	Male	Dappled	25
Manx	Female	Dappled	25
Did not survive development			25

Using suitable symbols, draw a genetic diagram to explain the results in Table 3.1.

[4]

- (b) A number of different crop plants have been genetically engineered to express a gene for an insecticidal toxin (*Bt* toxin) from a bacterium, *Bacillus thuringiensis*, that kills many insect species.

The *Bt* toxin binds to cell membranes of the cells in the insect gut. The receptor for the toxin has been identified as a membrane-anchored protein encoded by the cadherin gene. *Bt* toxin forms ion channels that cause an efflux (removal) of potassium ions from the insect gut cells, leading to cell lysis.

The cotton bollworm, *Helicoverpa armigera*, is one of the most serious insect pests of cotton in Asia, Australia, and Africa. A major deletion of the cadherin gene was associated with a high level of *Bt* toxin resistance in *H. armigera*. Three alleles (*r1*, *r2*, and *r3*) of the cadherin gene in the resistant strain have been identified:

- *r1* allele lacks 24-bp
- *r2* allele has a 202-bp deletion
- *r3* allele has a 126-bp deletion

- (i) Explain why mutations of the cadherin gene could confer resistance to *Bt* toxin in *H. armigera*.

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.....[2]

A survival test was conducted on larvae of different genotypes to assess their resistance to *Bt* toxin.

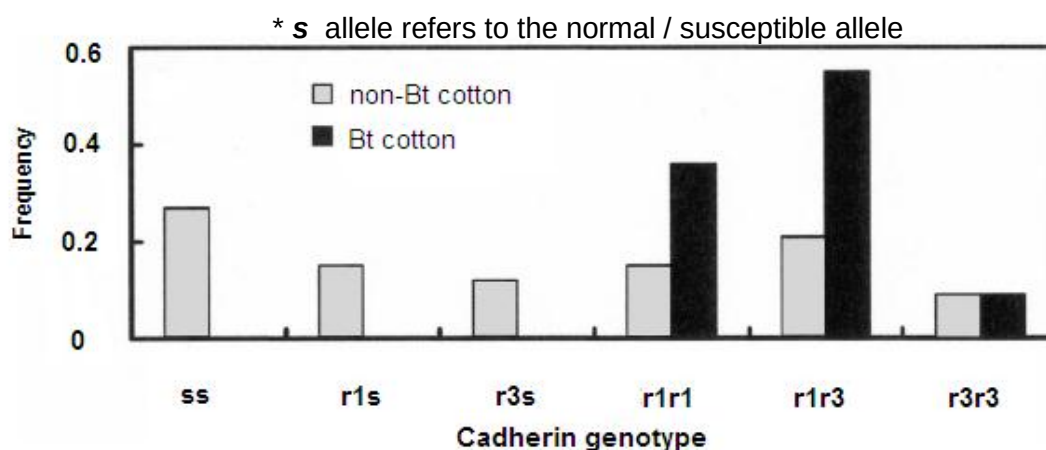


Fig. 3.1

- (ii) State and explain the nature of the mutant alleles that confer resistance to *Bt* toxin.

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.....[2]

Transgenic Atlantic salmon contains two foreign genes inserted into their genome.

Fig. 3.2 shows the gene construct of the two genes inserted into the Atlantic salmon; a promoter sequence from Ocean Pout and a growth hormone gene from the Chinook salmon.

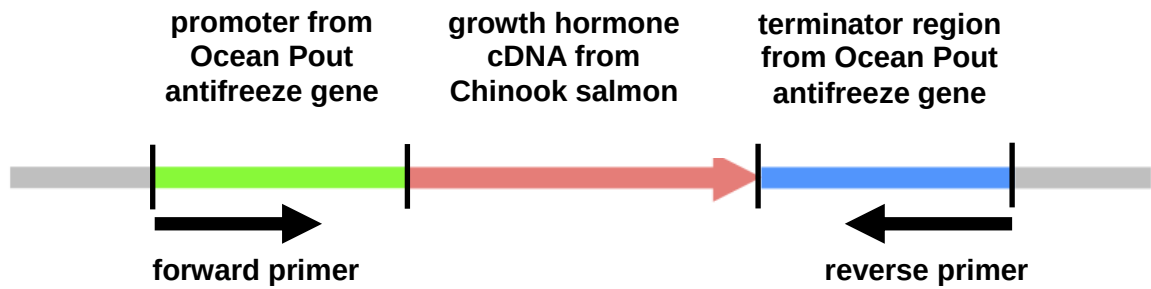


Fig. 3.2

PCR was carried out on all the samples to be investigated using the primers as shown in Fig 3.2.

(iii) With reference to Fig. 3.3, suggest why two different types of primers are needed.

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.....[2]

(iv) Explain how the two inserted genes might enable the transgenic salmon to meet the world's demand for food.

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.....[2]

[Total: 12]

- 4 (a)** A clinic sequenced the β -globin gene locus of five different patients and tabulated the results in the following table.

Table 1. Genetic profile of the 6th codon of patients' β -globin gene

Patient	DNA codon sequence*	Change in amino acid
1	GAG	Glu (unchanged)
2	GAA	Glu (unchanged)
3	GTG	Glu $\rightarrow$ Val
4	GAC	Glu $\rightarrow$ Asp
5	GTG	Glu $\rightarrow$ Val

* DNA sequence on the 6th codon of the human β -globin gene

Based on the information in Table 1,

- (i)** which patient(s) has/have the sickle cell anaemia mutation?

.....[1]

- (ii)** explain how the change of amino acid would result in a sickle shaped red blood cell.

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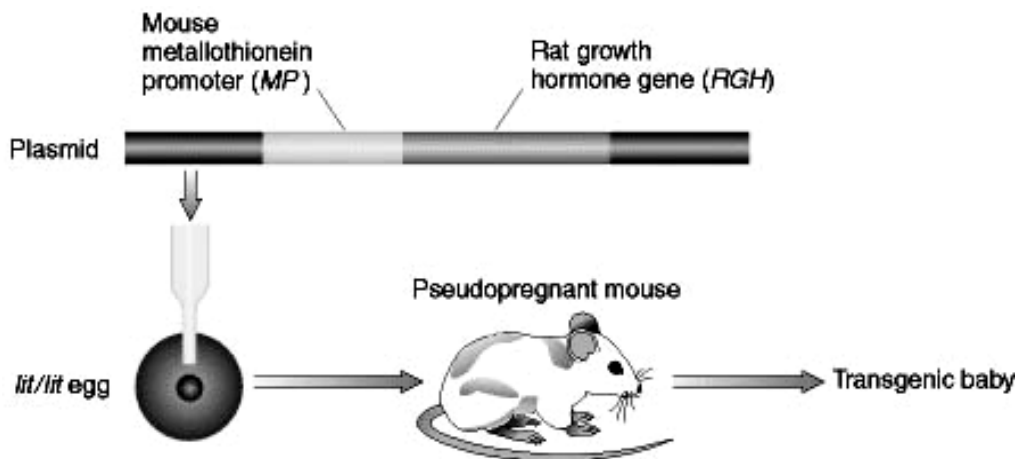
.....

.....[3]

- (b) The first example of gene therapy in a mammal was the correction of growth hormone deficiency in mice. The recessive mutation *little (lit)* results in dwarf mice. Even though a mouse's growth hormone gene is present, growth hormone is not produced.
- (i) Explain how the *little* mutation resulted in dwarfism.

.....[3]

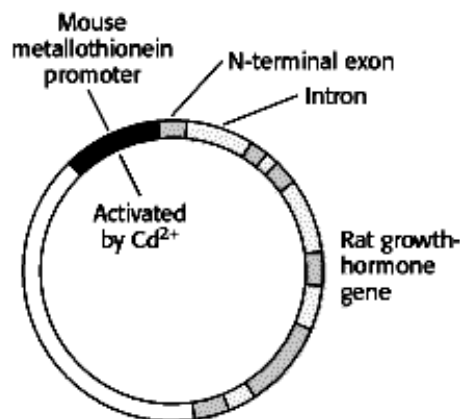
The initial step in correcting this deficiency was to inject homozygous *lit / lit* eggs with a 5-kb plasmid that contained the rat growth hormone structural gene (*RGH*) fused to a promoter sequence from the mouse metallothionein gene (*MP*). The normal role of the metallothionein protein is to detoxify heavy metals, hence the promoter sequence is responsive to the presence of heavy metals in the animal. The eggs were then implanted into pseudopregnant mice, and the transgenic baby mice were raised as shown in Fig. 4.1.



Source: Modern Genetic Analysis. Griffiths AJF, Gelbart WM, Miller JH, et al.
New York: W. H. Freeman; 1999.

Fig. 4.1

Fig. 4.2 shows the 5-kb plasmid that contained the *MP-RGH* gene construct. The *RGH* gene was inserted next to the *MP*, which is activated by the addition of heavy metals, such as cadmium ions to the drinking water.



Source: Biochemistry. 5th edition. Berg JM, Tymoczko JL, Stryer L.
New York: W H Freeman; 2002.

Fig. 4.2

(ii) Suggest the advantage of fusing the mouse metallothionein promoter (*MP*) to the rat growth hormone structural gene (*RGH*).

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.....[2]

(ii) Suggest why it is not necessary to include selectable markers in the plasmid.

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.....[2]

[Total: 11]

Section B

Answer **one** question.

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

5(a) Explain how one named factor affects enzymatic activity in relation to protein structure. [6]

5(b) Describe, with examples, the roles of membrane proteins in respiration [7]

5(c) Discuss how molecular techniques of DNA analysis can be used to provide evidence for molecular homology. [7]

[Total: 20]

OR

6(a) Explain how natural selection may bring about evolution. [7]

6(b) Explain the significance of triple helix in the structure and function of collagen. [6]

6(c) With reference to the term 'limiting factor', discuss how changes in light intensity affects the rate of photosynthesis. [7]

[Total: 20]

END OF PAPER